

The AI/ML World, Career Paths & Environment Setup

By the end of today you will understand what AI, ML, and Deep Learning actually mean, know which career path fits you best, and have your complete coding environment ready to use.

SECTION 01 What is AI, ML, and Deep Learning?

These three terms get mixed up everywhere — in the news, in job descriptions, in casual conversation. Let us fix that right now with clear, simple explanations.

The one idea that changes everything

When you were a child, nobody gave you a rulebook for recognising a dog. You were shown dogs — big ones, small ones, fluffy ones — and your brain slowly figured it out. That is the core idea behind all of AI and machine learning.

Old way: you write every single rule the computer must follow. ML way: you give examples and the computer finds the rules on its own.

Artificial Intelligence (AI) — the big umbrella

AI means making computers do things that normally need human intelligence — understanding language, recognising faces, making decisions. AI is the oldest and biggest category. It started in the 1950s. Machine Learning and Deep Learning both sit inside it.

Think of AI like the word "transport". A bicycle, a car, and an aeroplane are all transport. ML and Deep Learning are specific types of AI — just as cars and planes are specific types of transport.

Real-world examples of AI:

- A chess engine that beats grandmasters — uses rules and search, not learning
- Google Maps finding the fastest route to your destination
- Your phone's face unlock — recognises your face instantly
- ChatGPT and Claude answering questions in natural language

Machine Learning (ML) — AI that learns from data

ML is a specific approach to building AI. The computer is not given rules. Instead it is given data — thousands or millions of examples — and it finds the patterns itself. ML is the engine behind most practical AI you use today.

Everyday examples you already use:

- Gmail spam filter — learned from billions of spam vs. real emails
- Netflix recommendations — learned from what millions of viewers watched
- Your bank flagging a suspicious transaction — learned from fraud patterns
- Zomato delivery time estimate — learns from thousands of past deliveries
- Instagram Reels deciding what video to show you next

Analogy: Teaching a child to recognise apples. Old way: "An apple is round, red, has a stem." ML way: show the child 10,000 photos of apples and non-apples and let them figure it out. The ML child actually does better because it sees patterns a human cannot easily describe in words.

Deep Learning (DL) — ML using neural networks

Deep Learning is a type of ML that uses a structure called a neural network — loosely inspired by how the brain connects neurons. The word "deep" just means the network has many layers stacked on top of each other.

Deep Learning became powerful around 2012 when three things came together at once:

- Massive datasets — the internet gave us billions of labelled examples for free
- Cheap, powerful GPUs — graphics cards turned out to be perfect for this kind of math
- Better training techniques — researchers solved problems that had been blocking progress

Deep Learning is the technology behind everything that feels like magic today:

- ChatGPT and Claude — understanding and generating human language
- Midjourney — generating detailed images from a text description
- Face unlock on your phone — CNN (Convolutional Neural Network)
- Google Translate — real-time language translation across 100+ languages
- Tesla Autopilot — camera-based object detection and decision making

Important: Deep Learning needs far more data and computing power than regular ML. For many real business problems — spreadsheets, customer data, sales records — simpler ML models work just as well and are much easier to explain to a manager.

Data Science — the broader practice

Data Science is the skill of extracting useful insights and decisions from data. It uses statistics, programming, and business knowledge. It sometimes uses ML models, but not always.

A Data Scientist at a company might spend their day: pulling data with SQL, cleaning it in Python, making charts in Excel, and presenting a simple bar graph to management — no neural network in sight. That is still Data Science, and it is still valuable work.

Data Science is the job. AI and ML are tools you use inside that job. Most data science work is not glamorous model-building — it is data cleaning, exploring, and communicating findings clearly.

Quick comparison — all four terms side by side

Term	What it means in one line	Famous example
Artificial Intelligence	Machines doing human-like things	Chess engine, Siri, Alexa
Machine Learning	Machines learning patterns from data	Spam filter, Netflix recs
Deep Learning	ML using multi-layer neural networks	ChatGPT, face recognition
Data Science	Extracting insights from data	Business dashboards, A/B tests

SECTION 02 Five jobs you can get with this program

When you search "data science jobs" you will see many different titles. Here is what each one actually means day-to-day so you can pick the right target for yourself.

1. Data Analyst

Main question they answer: What happened, and why?

They pull data, clean it, build dashboards and charts, and answer business questions. Think of them as a reporter — they gather facts and present them clearly.

- Tools: SQL, Excel, Tableau, Power BI, basic Python
- A typical day: "Sales dropped 15% in Bengaluru last month. Find out why and make a chart."
- Entry salary in India (2024): ₹3.5 – 6 LPA

Great starting point for non-technical backgrounds. Has the most job openings of any data role.

2. ML Engineer

Main question they answer: How do we make this model work reliably for millions of users?

They build and deploy machine learning models. Half the job is engineering — packaging models so they run fast and do not crash when thousands of people use them simultaneously.

- Tools: Python, scikit-learn, TensorFlow/PyTorch, Docker, cloud platforms
- A typical day: "We have a trained fraud model. Now serve it as an API handling 5 million requests per day."
- Entry salary in India (2024): ₹6 – 12 LPA

This is the primary role this program is designed to train you for.

3. Data Scientist

Main question they answer: What will happen next, and what should we do about it?

A mix of statistics, machine learning, and business strategy. They run experiments, build predictive models, and present findings with confidence intervals and clear business recommendations.

- Tools: Python, R, statistics, ML libraries, Jupyter notebooks
- A typical day: "Which customers are likely to cancel next month? Build a model and tell marketing who to target."
- Entry salary in India (2024): ₹5 – 10 LPA

Needs the strongest foundation in statistics of any data role. If you love math and probability, this is your path.

4. AI Engineer

Main question they answer: How do we build a useful product powered by AI?

The fastest-growing role in 2024-2025. They build applications powered by large language models — chatbots, document summarisers, AI assistants. They often use existing AI APIs rather than training their own models from scratch.

- Tools: Python, Hugging Face, OpenAI/Anthropic APIs, RAG frameworks, Streamlit
- A typical day: "Build a chatbot that answers HR policy questions using our internal documents."
- Entry salary in India (2024): ₹7 – 14 LPA

The fastest growing role in the industry right now. Phase 5 of this program is entirely focused on this.

5. MLOps Engineer

Main question they answer: Why did the model stop working and how do we keep it healthy?

DevOps (keeping software running smoothly in production) applied specifically to ML models. They automate training pipelines, monitor models for accuracy drops, and retrain when performance degrades.

- Tools: Docker, Kubernetes, MLflow, CI/CD pipelines, AWS/GCP/Azure

- A typical day: "Fraud model accuracy dropped from 95% to 87% this week. Find out why, retrain, and automate weekly retraining."
- Entry salary in India (2024): ₹7 – 13 LPA

Usually requires 1-2 years of experience in ML engineering first. A natural step up from the ML Engineer role.

SECTION 03 AI you already use every single day

Before we write a single line of code, take a moment to see how much AI is already around you. Every one of these products runs on ML models built by people like you.

Product	What the ML model is doing
YouTube / Instagram Reels	Predicts which video keeps you watching longest and serves it next. Optimised for your time spent on the platform.
Gmail smart reply	When Gmail suggests "Sounds good!" — a language model read your email and predicted the most likely reply you would type.
Google Photos	Groups photos of the same person across thousands of images using a deep learning face-recognition model.
Amazon / Flipkart pricing	Product prices change thousands of times per day. ML models adjust based on demand, competitor prices, and your own browsing history.
Bank fraud alert	When you get an SMS saying "unusual activity" — an ML model flagged your transaction in under 100 milliseconds before any human saw it.
ChatGPT / Claude	A large language model trained on most of the internet. Predicts the most useful next word, millions of times per second, to construct a reply.

Every one of those products has a team of ML engineers and data scientists behind it. That is the kind of work this program is training you for.

SECTION 04 Setting up your environment

Follow the guide for your operating system. Every step is written out fully — nothing is skipped. If you get stuck at any point, screenshot the error and ask the trainer before moving forward.

Do you need Anaconda? No — not if you follow this guide. Anaconda is a bundle that installs Python + 250 libraries at once. It is heavy (3-4 GB) and can cause conflicts later. We install

Python directly and add libraries with pip — this is how real developers actually work. If you already have Anaconda installed, keep it. If you have nothing installed, start from Step 1 below.

Windows — Installation Guide

Step 1 — Install Python

1. Go to python.org/downloads

The site detects Windows and shows a yellow Download button. Click it. The file will be named something like python-3.12.x-amd64.exe.

2. Run the installer — this step is critical

Double-click the file. Before you click anything: look at the bottom of the first screen. Tick the checkbox that says "Add Python to PATH". This is the most important step on Windows — if you miss it, nothing will work from the terminal. Then click Install Now.

Missed the PATH checkbox? Uninstall Python from Control Panel > Add or Remove Programs, then reinstall and tick the box. Do not try to fix it manually.

3. Verify in Command Prompt

Press Windows key > type cmd > press Enter. In the black window, type:

```
python --version
```

Then also run:

```
pip --version
```

Both returned version numbers? Python is installed correctly and ready to use.

Step 2 — Install VS Code

4. Download from code.visualstudio.com — click Download for Windows
5. Run the installer. On the "Select Additional Tasks" screen tick: Add 'Open with Code' to file context menu, and Add to PATH. Then click Install.
6. Open VS Code. Click the Extensions icon (4 squares on left sidebar). Search Python and install the one by Microsoft. Then search Jupyter and install that too.
7. Press Ctrl + Shift + P > type Python: Select Interpreter > choose Python 3.x.x.
8. Test: File > New File > save as hello.py > type `print("Hello, World!")` > click the Run button (triangle top-right). You should see Hello, World! in the terminal below.

Saw Hello, World! in the terminal? VS Code and Python are connected and working correctly.

Step 3 — Install Git and set up GitHub

9. Download Git from git-scm.com/downloads. Run the installer keeping all default options.
10. Close and reopen Command Prompt. Verify: type `git --version` and confirm you see a version number.
11. Configure your identity — run both commands:

```
git config --global user.name "Your Full Name"
git config --global user.email "youremail@gmail.com"
```

12. Go to github.com > Sign up with a professional username (no nicknames — this goes on your resume).
13. On GitHub: click + > New repository > name it ai-ml-bootcamp > tick Add a README > Create. Click the README.md file > pencil icon to edit > add your name, background, and why you joined > Commit changes.

README showing your text on GitHub? Git and GitHub are fully set up.

Step 4 — Install core Python libraries

Open Command Prompt and run:

```
pip install numpy pandas matplotlib seaborn scikit-learn jupyter
```

Final test — create test.py in VS Code and run it:

```
import numpy as np
import pandas as pd
print("All libraries loaded. We are ready!")
```

Saw the message? Windows setup is 100% complete.

Mac — Installation Guide

Important: Macs come with an old Python 2.7 pre-installed. Do not use that one — it is outdated. We need Python 3 from the official website. Also note: on Mac you always use `python3` and `pip3`, not `python` or `pip` alone.

Step 1 — Install Python

14. Go to python.org/downloads. The site detects Mac and shows a .pkg installer file. Click Download.
15. Double-click the .pkg file > click Continue through all screens > click Install. It will ask for your Mac password — this is normal.
16. Open Terminal: press Cmd + Space > type Terminal > press Enter. Verify:

```
python3 --version
pip3 --version
```

Both show version numbers? Python 3 is installed correctly on your Mac.

Step 2 — Install VS Code

17. Download from code.visualstudio.com. You get a .zip file. Double-click to unzip — it produces Visual Studio Code.app.
18. Drag Visual Studio Code.app to your Applications folder. Do not run it from Downloads.
19. Open VS Code > press Cmd + Shift + P > type shell command > click Shell Command: Install code command in PATH. Now you can open any folder in VS Code from Terminal by typing code .
20. Install extensions: click Extensions icon (4 squares) > search Python > install by Microsoft. Then search Jupyter > install by Microsoft.
21. Press Cmd + Shift + P > Python: Select Interpreter > choose Python 3.x.x.
22. Test: create hello.py > type print("Hello, World!") > click Run. Confirm you see the output.

Saw Hello, World!? VS Code is connected to Python correctly.

Step 3 — Install Git and set up GitHub

First check if Git is already installed. In Terminal:

```
git --version
```

If you see a version number — skip to configuring your identity. If you see command not found, run:

```
xcode-select --install
```

A popup appears — click Install. This takes 5-10 minutes and includes Git. After it finishes, configure your identity:

```
git config --global user.name "Your Full Name"
```

```
git config --global user.email "youremail@gmail.com"
```

Create your GitHub account at github.com with a professional username. Create your repo ai-ml-bootcamp with a README. Edit and commit your introduction.

Step 4 — Install core Python libraries

On Mac use pip3:

```
pip3 install numpy pandas matplotlib seaborn scikit-learn jupyter
```

Final test — same as Windows. Create test.py, run the import statements, confirm you see the success message.

Saw the message? Mac setup is 100% complete.

Linux (Ubuntu / Debian) — Installation Guide

These steps are for Ubuntu and Debian-based distributions. If you use Fedora or Arch, the package manager commands differ but the process is identical.

Step 1 — Install Python

First check what is already there. Open Terminal (Ctrl + Alt + T):

```
python3 --version
```

If you see Python 3.x.x — skip to Step 2. If not, install it:

```
sudo apt update
```

```
sudo apt install python3 python3-pip -y
```

sudo means run as administrator. It will ask for your password. Type it — you won't see any characters appearing as you type. That is normal on Linux.

Verify both:

```
python3 --version
```

```
pip3 --version
```

Both show version numbers? Python is ready.

Step 2 — Install VS Code

23. Download from code.visualstudio.com > Download for Linux > choose the .deb file.

24. In Terminal:

```
cd ~/Downloads
```

```
sudo apt install ./code_*.deb
```

25. Launch VS Code by typing code in Terminal or finding it in Applications. Install the Python and Jupyter extensions as described above.

26. Press Ctrl + Shift + P > Python: Select Interpreter > choose Python 3.

27. Test with hello.py and confirm Hello, World! appears.

Step 3 — Install Git and set up GitHub

Check first:

```
git --version
```

If not found, install:

```
sudo apt install git -y
```

Configure identity:

```
git config --global user.name "Your Full Name"
```

```
git config --global user.email "youreemail@gmail.com"
```

Create GitHub account and ai-ml-bootcamp repo as described above.

Step 4 — Install core Python libraries

```
pip3 install numpy pandas matplotlib seaborn scikit-learn jupyter
```

Getting an "externally managed environment" error? Add the flag: pip3 install numpy pandas matplotlib seaborn scikit-learn jupyter --break-system-packages

Run the final test (import numpy, pandas, print success message) to confirm everything works.

Saw the message? Linux setup is 100% complete.

Google Colab — Works on Every Device, Zero Installation

If you have a slow laptop, a Chromebook, a tablet, or anything that can run a browser — Colab is your answer. It runs entirely in the cloud. Python, all the data science libraries, and even a free GPU are all provided without installing a thing.

28. Go to colab.research.google.com and sign in with your Gmail account.
29. Click New notebook. A Jupyter notebook opens in your browser. Every data science library is already installed — numpy, pandas, matplotlib, everything.
30. To get a free GPU for Phase 3: Runtime > Change runtime type > T4 GPU > Save.
31. Notebooks auto-save to Google Drive > My Drive > Colab Notebooks.

Colab is used by researchers at Google, Stanford, and top AI companies every single day. Using it is not a shortcut — it is smart tool selection.

SECTION 05 How ML actually works — the big picture

You will build every one of these steps over the next 28 days. For now, just understand the overall flow and how the pieces connect.

- 1 Define the problem**
What exactly are we trying to predict? "Will this customer leave next month?" is a clear problem. "Do something with data" is not. The clearer your question, the better your model will be.
- 2 Collect data**
Get raw data — a CSV file, a database, an API feed. The quality of your data decides 80% of how well your model works. Garbage in = garbage out.
- 3 Explore and clean the data (EDA)**
Fix missing values, remove duplicates, understand what each column means, find unusual values. This is called Exploratory Data Analysis. Days 7-12 of this program are focused here.
- 4 Prepare features**
Transform raw columns into things the model can learn from. A date-of-birth becomes an age. A city name becomes a number. This is called feature engineering — the most creative part of ML.

5

Train the model

Feed the prepared data into a learning algorithm. It adjusts thousands of internal numbers until it can make accurate predictions on data it has not seen before.

6

Evaluate the model

Test it on data it has never seen. Is it accurate? Is it fair? Does it work on unusual cases? Numbers alone are not enough — understand why it makes each prediction.

7

Deploy and monitor

Make the model available as an app or API so real people can use it. Then watch it over time — the real world changes, and models need retraining to stay accurate.

Full analogy: Training a bank employee to spot fake currency. You show them real and fake notes (data). They study patterns (training). You test them on new notes (evaluation). They pass and sit at the counter (deployment). You check their accuracy monthly (monitoring).

SECTION 06 Interview prep — questions to answer after today

After every class you will see questions an interviewer might ask on this topic. Practice saying the answer out loud — not just reading it in your head.

Q — What is the difference between AI and Machine Learning?

AI is the broad goal of making machines behave intelligently. Machine Learning is one specific approach to achieving that goal — where the machine learns patterns from data rather than following hand-coded rules. All ML is AI, but not all AI is ML. A chess engine that uses search rules is AI but not ML. Gmail's spam filter that learns from examples is both AI and ML.

Q — Can you give a real business example of ML being used?

Credit card fraud detection is a strong example. Banks train a model on millions of past transactions labelled as fraud or not fraud. The model learns which patterns — unusual location, unusual amount, unusual time of day — are suspicious. When a new transaction comes in, the model scores it in under 100 milliseconds. Writing individual rules to cover every fraud scenario would be impossible.

Q — What is Deep Learning and when would you use it instead of regular ML?

Deep Learning is a type of ML that uses neural networks with many layers. It works best for unstructured data — images, audio, and text — where it is hard to manually describe what features to look for. For structured data like spreadsheets and tables, simpler ML models often work just as well and are much easier to explain to non-technical stakeholders. The choice depends on data type, data volume, and whether explainability is required.

SECTION 07 Words to know

These terms will come up constantly from tomorrow onwards. Know them cold.

Term	What it means
Algorithm	A set of instructions. In ML, a recipe for finding patterns in data.
Model	The result of training an algorithm on data. The "brain" that makes predictions.
Feature	A column in your data. An input the model uses — such as age, salary, or city.
Label / Target	The answer you are trying to predict — fraud yes/no, house price, customer churn.
Training data	The examples you show the model so it can learn — like the textbook a student studies.
Repository (repo)	A GitHub project folder. One project = one repo. Your portfolio = all your repos.
Commit	A saved snapshot of your code with a message explaining what changed.
Deploy	Making your model or app available for real users to access online.

SECTION HW Homework — submit before Day 2

These are not optional. Daily exercise completion is graded pass/fail. Three missed submissions = formal warning.

32. Verify all four tools work — Python, VS Code (with Python and Jupyter extensions), Git, and GitHub. If any one is broken, fix it tonight.
33. Create your GitHub account if you have not already. Create the repo ai-ml-bootcamp with a README file.
34. Edit the README with: your name, your background (be honest — everyone starts somewhere), and 2-3 sentences on why you joined this program. Commit and push it.
35. Run `print("Hello, World!")` in three separate places: VS Code terminal, a Jupyter notebook cell, and Google Colab. Screenshot all three.
36. Submit your GitHub repo link to the trainer before midnight tonight.

No GitHub link submitted = incomplete for Day 1. This is your first portfolio piece — take it seriously.

